

# ToothPrint: A Certified, Partial-Overlap-Robust System for Dental Biometric Identification

Krishi Attri

Independent research | [kattri@snu.ac.kr](mailto:kattri@snu.ac.kr) | [github.com/Archerkattri/toothprint](https://github.com/Archerkattri/toothprint)  
Methods report. Every number is reproduced by a committed script + result JSON; every limitation is stated.

## Abstract

The dental arch is an individual “tooth print”: crown contours, cusp geometry, the gingival margin, and the pattern of restorations. We assemble a dental-identity system that (i) recognises a person from a 3D intraoral scan or a 2D radiograph, (ii) returns a **certified** accept/abstain decision with a distribution-free false-match-rate bound, and (iii) is **robust to partial overlap**—the missing-teeth query where prior rigid methods collapse. Our contributions are a learned point-correspondence matcher that lifts 50%-tooth-loss identification from 0.23 (rigid GICP) to **0.87**, a conformal open-set decision hardened against look-alikes, a restoration biometric demonstrated on both CBCT and plain radiographs, and an honest characterisation: full DET curves, ablations, a cross-dataset domain-gap measurement, and the regimes where the system must abstain. The one limit no method here closes is real cross-session data.

## 1 Introduction

Forensic and clinical dental identification is classically manual (charting) or, recently, rigid 3D scan registration. Two gaps persist: (a) no *certificate*—methods report Rank-N accuracy but never a finite-sample bound on the false-match rate of an individual verdict; and (b) *partial overlap*—a query arch missing teeth (extraction, trauma, field-of-view) breaks the rigid alignment that all registration methods depend on. ToothPrint targets both, and treats honesty about failure regimes as a first-class deliverable.

## 2 Related work and positioning

The only dedicated 3D-intraoral identity pipeline with proper Rank-1 is Zhou et al. 2024 (*Bioengineering*)—FPFH + SAC-IA + ICP + RMSE, the same registration family as our geometric core—reporting Rank-1 100% on 160 real adults with  $\sim 1$ -year re-scans (closed-set, private data). We match its saturated accuracy (Rank-1 0.995 / EER 0.5% on 200 arches) and **lead on a certified, bounded-FMR decision with open-set rejection** that no prior dental work reports, and on **learned partial-overlap correspondence**, which the registration literature lacks for this domain. ToothPrint’s defensible lane is *certification + partial-overlap robustness*, not a higher saturated number.

## 3 Methods

**Geometric identity (3D).** A query re-scan is given its best *rigid* fit to each gallery arch—PCA principal-axis initialisation (four proper-rotation hypotheses; robust to

the self-similar palate that defeats FPFH) refined by multi-scale Generalised-ICP—and scored by mean point-to-surface distance. Rigid (no scale) so the score is shape, not pose.

**Learned embedding.** A DGCNN (EdgeConv) encoder with a sub-centre ArcFace head maps an arch to an L2-normalised descriptor. *Crop-hardening*: training with aggressive partial crops (keep  $\geq 0.35$ ) makes the descriptor coverage-robust at no full-coverage cost.

**Learned point correspondence (CorrNet).** The key partial-overlap result. A DGCNN backbone emits a **unit descriptor per point**, trained with an InfoNCE correspondence loss—cropping a canonical point set yields ground-truth point matches for free. At test, a partial query is matched point-to-point against each gallery arch (mutual nearest-neighbours  $\rightarrow$  weighted Procrustes), and the residual over *all* query points scores the fit.

**Conformal certificate.** Calibrating a threshold on held-out genuine / nearest-impostor scores yields an empirical false-match rate  $\leq \alpha$ , distribution-free and finite-sample.

**Unified decision.** Retrieve by embedding (recall)  $\rightarrow$  verify the shortlist by CorrNet correspondence (precision)  $\rightarrow$  accept above the conformal threshold, else **abstain**.

**Multimodal & dental-work.** On paired CBCT+IOS, three independent biometrics—IOS crowns, CBCT bone/root geometry, CBCT restoration cloud (HU  $> 2500$ )—are scored and fused. The restoration pattern also extends to 2D radiographs via a per-tooth local-contrast extractor (a restoration is the patch far brighter than its own tooth’s median).

| method                             | keep-0.5    | keep-0.3    |
|------------------------------------|-------------|-------------|
| rigid GICP                         | 0.23        | 0.10        |
| crop-hardened embedding            | 0.64        | 0.26        |
| <b>CorrNet (realistic dropout)</b> | <b>0.87</b> | <b>0.57</b> |
| CorrNet (planar crop)              | 0.91        | 0.80        |

Table 1: Rank-1 under tooth loss, held-out unseen subjects.

## 4 Datasets and protocols

Poseidon3D (intraoral STL, 200: 150 train / 50 held-out), Teeth3DS+ (intraoral OBJ, 150, cross-dataset), DenPAR (periapical radiographs, 400 / 165 restoration-bearing), Figshare CBCT+IOS (paired, 55). All are single-timepoint, so **genuine queries are synthetic**—acquisition repositioning, sensor noise, and partial crops (planar or realistic whole-tooth dropout). This is disclosed on every result and is the central limitation (§7).

## 5 Results

**Identity (full coverage).** 3D Rank-1 **0.995** (N=200, EER 0.5%, AUC 0.997, fidelity 0.05mm); 2D Rank-1 **1.000** (N=400, EER 0). Conformal empirical FMR tracks  $\alpha$  in every ablation. Fig. 2 gives full DET curves—a first for dental identity.

**Partial overlap (the contribution).** Table 1 and Fig. 1 (top-left): under realistic whole-tooth dropout, CorrNet lifts keep-0.5 from 0.23 (GICP) to **0.87**. An *untrained* CorrNet already scores 0.70—so the gain is  $\sim$ half the correspondence-plus-rigid-verification *architecture* and  $\sim$ half the *learned* descriptors. Honest correction kept: planar-crop numbers were optimistic vs realistic dropout (keep-0.3 0.80 $\rightarrow$ 0.57).

**Multimodal fusion.** On 55 paired patients: IOS 1.000, CBCT-bone 0.945, dental-work 0.927. In a degraded regime the **oracle bound is 1.000** (modalities complementary), naive equal-weight fusion *hurts* (dilution), and quality-weighted fusion beats the best single (0.867 vs 0.833)—a real but Rank-1-only gain (its AUC regresses).

**Dental-work on 2D radiographs.** Per-tooth local contrast (global thresholding fails on over-saturated JPEGs) identifies 165 subjects at Rank-1 **0.91–0.99** (robust to jitter + a dropped restoration; chance 0.006).

**Unified certified decision.** Full coverage FNIR@FMR=1% = 0.00; partial (keep-0.5) 0.74—best of all methods but does not escape the open-set ceiling (§6), so it abstains.

## 6 Honest limitations

(1) **Open-set collapses under partial overlap.** At keep-0.5 every method’s FNIR@FMR=1% is 0.74–0.87

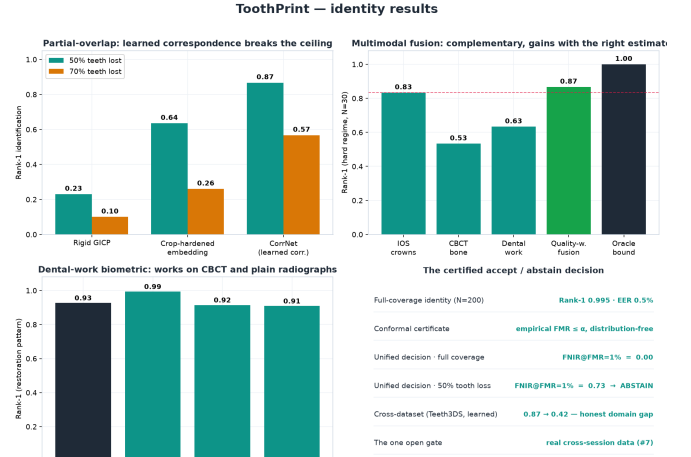


Figure 1: Headline results: partial-overlap breakthrough, fusion complementarity, dental-work across modalities, the certified decision.

(vs 0.03 full)—a half-arch fits many gallery arches, so impostor/genuine score tails overlap. Certified rejection is a near-full-coverage property; under heavy tooth loss the correct behaviour is to **abstain**. (2) **Cross-dataset domain gap.** CorrNet trained on Poseidon3D, run on Teeth3DS+, drops keep-0.5 0.87 $\rightarrow$ 0.42 (still 34 $\times$  chance, still >GICP); the learned descriptors are partly dataset-specific. (3) **Extreme tooth loss** (keep-0.3) is the hard tail (0.57 realistic). (4) **Fusion gain is Rank-1-only and small** (AUC regresses under naive weighting).

## 7 The binding constraint: real cross-session data

Every identity number here—CorrNet’s 0.87, the 0.995, the Teeth3DS results—rests on **synthetic** re-scans/crops of single-timepoint arches, because no open dataset pairs the same person across visits with the needed modality. This is a *data* limit no code closes. Until such data validates the pipeline, ToothPrint is **best-by-design and validated in simulation**, not proven on real longitudinal data—and the repository states this throughout.

## 8 Reproducibility

Datasets are gitignored; paths are env-configurable, reference baselines are read from committed artifacts, and committed synthetic fixtures run the 3D pipeline end-to-end with no off-machine data (TOOTHPRINT\_FIXTURES=1; see REPRODUCE.md).

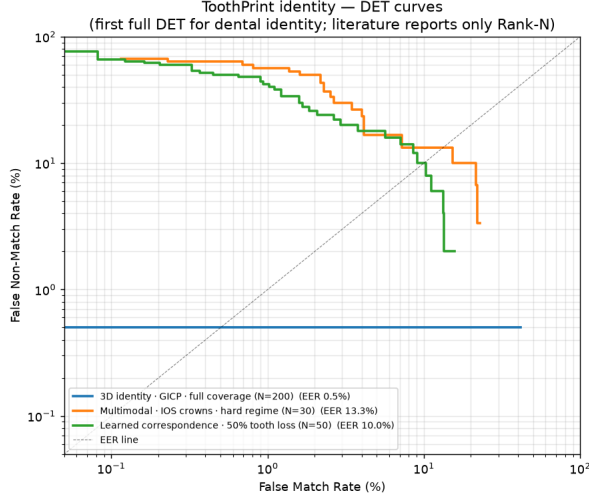


Figure 2: Full DET curves per identity pillar (first for dental identity).

## 9 Conclusion

ToothPrint is, to our knowledge, the most complete and most partial-overlap-robust **certified** dental-identity system demonstrable without gated longitudinal data: a learned correspondence matcher that quadruples rigid performance under 50% tooth loss, a conformal accept/abstain certificate, a dental-work biometric across CBCT and radiographs, and a discipline of reporting the regimes where it must decline. The path to a clinical claim runs through real cross-session data, not more code.

**References.** Zhou et al. (2024), *Bioengineering*—3D intraoral-scan identification. Wang et al. (2019)—Dynamic Graph CNN (EdgeConv). Deng et al. (2019/2020)—(Sub-centre) ArcFace. Segal et al. (2009)—Generalised-ICP. Vovk et al.—Conformal prediction. Qin et al. (2022)—GeoTransformer.